Project Summary

Design and Development of an Upper Limb Rehabilitation Exoskeleton for the Elbow and Wrist

Author: Ansh Duggal

University of Leeds

BEng (Hons) Mechatronics and Robotics

Email: anshduggal10@gmail.com

This project presents the design and development of a two-degree-of-freedom (DOF) rehabilitation exoskeleton for assisting elbow and wrist flexion-extension movements in stroke patients. The goal was to create a lightweight, affordable, and user-friendly device for at-home rehabilitation without continuous therapist supervision.

The mechanical design was modelled using SolidWorks and fabricated using 3D-printed ABS material, balancing strength, comfort, and low cost. The elbow joint is actuated by a NEMA 23 stepper motor delivering 1.26 Nm torque and a 1.8 degrees step angle, while the wrist joint uses two BMS-620MG servo motors providing 140 Ncm torque for smooth and precise control.

An Arduino UNO R4 WiFi microcontroller with IoT Cloud integration enabled wireless control through a web or mobile dashboard. The system allows joint movement via sliders and buttons, along with real-time feedback from an MPU-6050 gyroscope and LCD display for monitoring orientation and motion data.

Testing validated smooth motion, consistent torque transfer through a flange coupler, and reliable IoT control. The prototype achieved full elbow motion (0 degrees-135 degrees) and functional wrist assistance.

Key Achievements:

- Fully functional prototype within a £100 budget.
- Successful wireless IoT-based control and monitoring system.
- Modular, lightweight 3D-printed design ensuring comfort and adjustability.
- Integrated sensors for motion tracking and safety.

Future improvements include reducing segment weight, enhancing user comfort through softer straps, and refining connectivity for commercial scalability.